

BeeWare and Kivy: Python Tools for Building Cross-Platform Apps

The ability to build apps is an important skill for a software developer. Building applications that run across multiple platforms, such as both Android and iOS, is important today. This article explores two tools in Python that help to build cross-platform apps—BeeWare and Kivy.

Apps are everywhere. One of the major reasons for the popularity of devices such as smartphones is the availability of innumerable apps. The ability to build apps is an important skill that software developers aim to acquire. As there are many platforms these days such as Android, iOS, Raspberry Pi, etc, developing applications that run across all of these has many benefits like consistency, singular code, etc. Hence, building cross-platform applications is getting popular among developers.

There are many languages to select from when you are developing an app. For example, if you want to develop an Android app, you have the following options:

- Java
- Kotlin
- Python
- C / C++ / C#

Apart from these, you can use Web technologies such as HTML, CSS and JavaScript as well to build mobile apps. The languages listed above cover only one platform — Android. Likewise, there are language options

for each of the platforms such as Raspberry Pi and iOS.

Selecting a particular language for development depends on various factors such as the nature of the application, the target user base and the expertise available. As per the *IEEE Spectrum* report, Python topped the language rankings list in 2019. Due to the availability of a large number of Python developers and the ease with which this language can be learned, it is a very good choice for app development with specialised frameworks and libraries.

This article focuses on building cross-platform apps. For this purpose, there are two important options in Python, as listed below:

- Kivy
- BeeWare

Both these frameworks have certain commonalities and differences as well. This article provides an introduction to their features and the steps involved in building your first app using these frameworks.

Kivy

Kivy is an open source Python library. It can be used for the rapid development of apps. It supports features such as multi-touch. The major features of Kivy are listed below:



Figure 1: Industry sectors and applications of AI, ML and DL

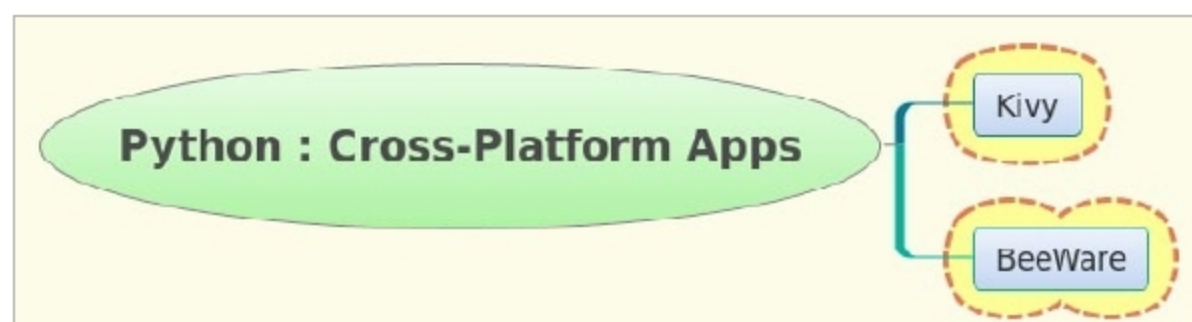


Figure 2: Cross-platform apps with Python

